

AgroAssist AI

Intelligent Agricultural Co-Pilot for Rwandan Smallholder Farmers

Startup Presentation 2026

1. THE SCALE OF THE PROBLEM — BY THE NUMBERS

Rwanda has over 2 million smallholder farming households. Yet the agricultural support infrastructure has not kept pace. The numbers below show exactly why AgroAssist AI exists:

2M+

Farmers in Rwanda
smallholder households

1:1,000+

Agronomist Ratio
farmers per agronomist

**7–14
days**

Avg. Diagnosis Wait
before AgroAssist

~30%

Annual Crop Loss
from preventable
disease

🕒 The 5-Minute Opportunity: When a farmer sees a diseased leaf today, the clock starts. It takes 2–14 days on average to get a diagnosis — by then, the crop may be lost. AgroAssist AI delivers a diagnosis in under 30 seconds. That single shift in speed is the foundation of everything this product does.

2. WHY FARMERS ARE LOSING — PAIN POINTS IN DETAIL

Each problem below is a direct threat to a farmer's livelihood. These are not inconveniences — they are the difference between a profitable harvest and a lost season.

1 Diagnostic Delay

There is roughly 1 agronomist for every 1,000+ farmers in Rwanda. Getting a disease identified for Kinigi Potatoes or Maize can take 1–2 weeks — during which the blight spreads. On average, this delay contributes to ~30% annual crop losses that are entirely preventable.

② Input Misuse & Wasted Capital

Without proper guidance, farmers purchase the wrong pesticides — sometimes counterfeit, sometimes completely ineffective. Studies in Sub-Saharan Africa show up to 40% of smallholder pesticide purchases are misapplied, burning cash that families cannot afford to lose.

③ Information Asymmetry at Market

Farmers sell produce without knowing the real Kigali wholesale or retail price. Middlemen exploit this gap. A farmer receiving 200 RWF/kg for potatoes that sell for 350–400 RWF in Kigali markets loses 30–40% of potential revenue per harvest cycle.

④ Weather-Induced Spray Waste

A farmer can spend 5,000–15,000 RWF on pesticides only to have rain wash them off the same day. With no weather-integrated spray advisory, this is a routine loss.

⑤ Connectivity Gaps in the Field

High-density farming zones — Musanze, Rubavu, Nyamagabe — regularly experience poor or absent cellular data. Any cloud-only solution fails these farmers at the most critical moment.

3. WHAT WE ARE GIVING — BEFORE VS. AFTER AGROASSIST AI

The table below maps each pain point to the specific tool that resolves it, and what the farmer gains in real terms:

TASK / CHALLENGE	BEFORE AgroAssist	WITH AgroAssist AI
Identify crop disease	1–14 days (wait for agronomist)	< 30 seconds (AI photo scan)
Get treatment guidance	Travel to district office	Instant, on-device recommendation
Check pesticide safety	Trial & error / peer advice	Rwanda MOA approved list, instantly
Verify market price	Accept middleman's quote	Live price index, negotiation power

Safe-to-spray check	No system — guess or lose money	Integrated weather alert before spray
Work without internet	Impossible (cloud-only tools)	Offline queue auto-syncs when back online

4. THE 5-IN-1 TOOLSET — FEATURE OVERVIEW

Feature	What It Does	Coverage	Farmer Gain
AI Crop Scan	Photo → Diagnosis in < 30s	Maize, Potatoes, cash crops	Weeks → Seconds
Medicine DB	MOA-approved pesticide list	Natural + chemical options	Zero wrong purchases
Offline-First	Queue scans, auto-sync later	Works in rural fields	100% uptime in field
Market Intel	Live Kigali price index	Potato, Maize benchmarks	+10–15% revenue
Spray Safety	Weather-based spray alert	Prevents wash-off waste	–15% input costs

5. WHAT WE EXPECT — MEASURABLE IMPACT

Success is not measured in features shipped — it is measured in farmer outcomes. These are the targets we are accountable to:

Metric	Target	How AgroAssist Delivers It
Yield Retention	+20–30%	Early blight detection stops total harvest failure
Input Cost Savings	–15%	Right pesticide + weather check = no waste

Farmer Revenue Uplift	+10–15%	Market data removes middleman exploitation
Diagnosis Time	Weeks → 30 sec	From waiting for agronomist to instant AI answer
Connectivity Resilience	100% uptime	Offline mode queues & syncs — no data lost

6. ROADMAP — WHERE WE ARE & WHERE WE ARE GOING

Phase 1: Core Build DONE	Phase 2: Pilot ACTIVE	Phase 3: Launch UPCOMING
- UX/UI Optimization (May 20 – Jun 5, 2026) - Market Intel Fallback (Jun 6 – 8, 2026)	- Beta Guide Distribution → pilot farmers (Jun 9–15) - Field testing in Musanze & Kigali (Jun 16 – Jul 10)	- Production Web Deploy to Firebase (Jul 11–25) - Google Play Store Launch (Jul 26 – Aug 10)

Current status: Beta v1.2 APK is built and ready for field distribution. We are targeting 20-50 pilot farmers across Rwanda.

AgroAssist AI — Reducing Diagnosis Time from Days to Seconds.
Making every farmer's 5 minutes count.